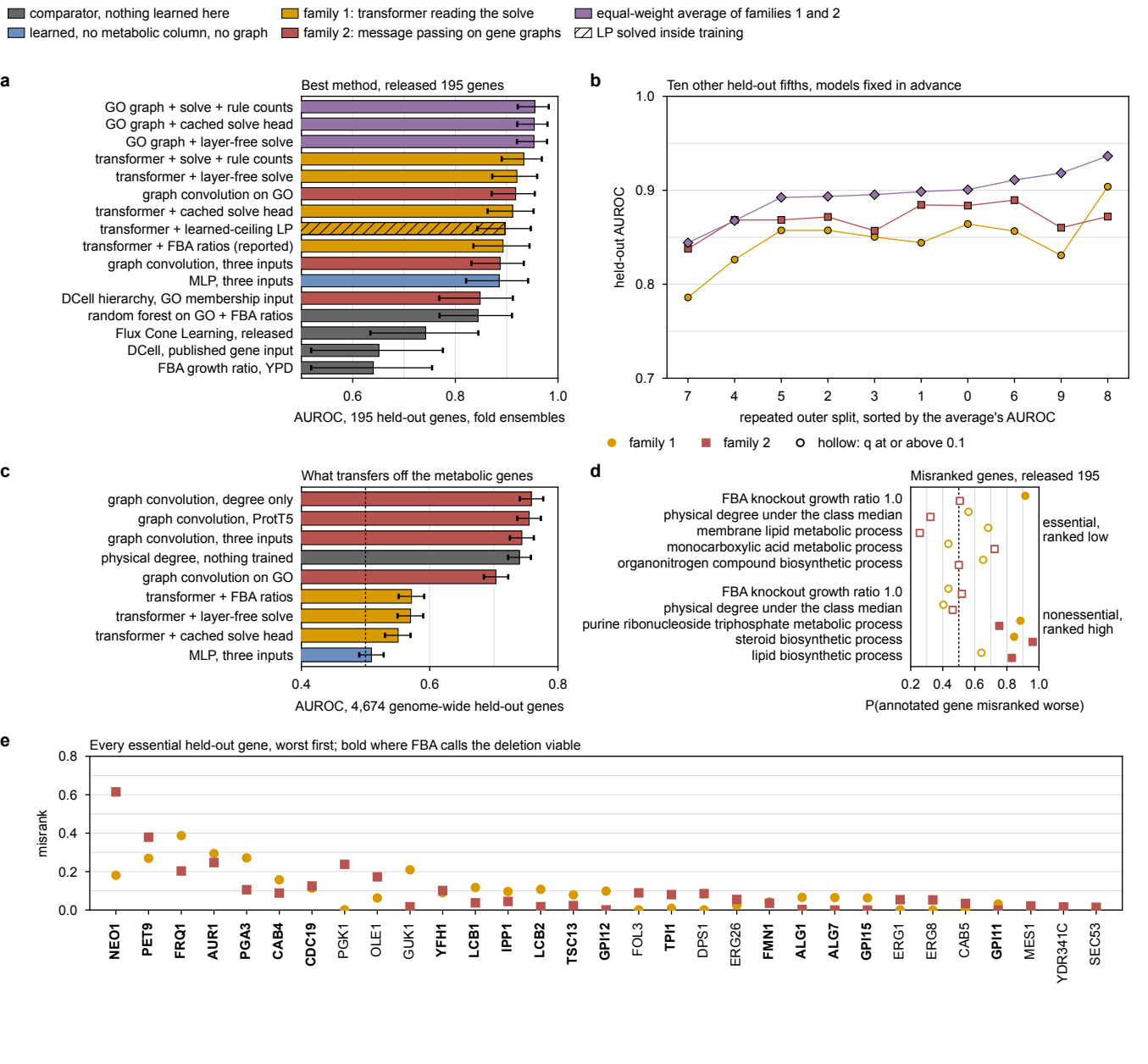
comparator, nothing learned here
learned, no metabolic column, no graph
family 1: transformer reading the solve
family 2: message passing on gene graphs
equal-weight average of families 1 and 2
LP solved inside training
a
Best method, released 195 genes
GO graph + solve + rule counts
GO graph + cached solve head
GO graph + layer-free solve
transformer + solve + rule counts
transformer + layer-free solve
graph convolution on GO
transformer + cached solve head
transformer + learned-ceiling LP
transformer + FBA ratios (reported)
graph convolution, three inputs
MLP, three inputs
DCell hierarchy, GO membership input
random forest on GO + FBA ratios
Flux Cone Learning, released
DCell, published gene input
FBA growth ratio, YPD
0.6
0.8
1.0
AUROC, 195 held-out genes, fold ensembles
b
Ten other held-out fifths, models fixed in advance
held-out AUROC
1.0
0.9
0.8
0.7
7 4 5 2 3 1 0 6 9 8
repeated outer split, sorted by the average's AUROC
family 1
family 2
hollow: q at or above 0.1
c
What transfers off the metabolic genes
graph convolution, degree only
graph convolution, ProtT5
graph convolution, three inputs
physical degree, nothing trained
graph convolution on GO
transformer + FBA ratios
transformer + layer-free solve
transformer + cached solve head
MLP, three inputs
0.4
0.6
0.8
AUROC, 4,674 genome-wide held-out genes
d
Misranked genes, released 195
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
essential, ranked low
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
nonessential, ranked high
0.2 0.4 0.6 0.8 1.0
P(annotated gene misranked worse)
e
Every essential held-out gene, worst first; bold where FBA calls the deletion viable
misrank
0.8
0.6
0.4
0.2
0.0
NEO1 PET9 FRQ1 AUR1 PGA3 CAB4 CDC19 PGK1 OLE1 GUK1 YFH1 LCB1 IPP1 LCB2 TSC13 GPI12 FOL3 TPI1 DPS1 ERG26 FMN1 ALG1 ALG7 GPI15 ERG1 ERG8 CAB5 GPI11 MES1 YDR341C SEC53